TECHNICAL DESIGN DOCUMENT (TDD)

DumHouse: Direct Food-Ordering Platform with Real-Time Kitchen Display System (KDS) & AI Preparation-Time Estimator

Project Metadata & Review Links

- **GitHub Project Repository:** `https://github.com/naman-isocode/dumhouse-platform` *(Place actual repository URL here before submission)*
- **Live Production Application:** `https://dumhouse-platform.vercel.app` *(Place production Vercel/Supabase URL here)*
- **Client Organization:** Company A, Hyderabad (Biryani & Authentic Hyderabadi Food Chain)
- **Scale:** 14 Outlets, 3 Cloud Kitchens, 520 Staff, ■90–105 Crore Annual Revenue
- **Prepared For:** Isocode Software Engineering Take-Home Assessment (Problem Statement 3)
- **Author:** Software Engineering Candidate

1. Executive Summary & Business Rationale

1.1 The Client's Dilemma

Company A is a premier Hyderabad restaurant chain celebrated for authentic Handi Dum Biryani, Mutton Haleem, and Nizami specialties. Despite generating ■90–105 crore in annual revenue, over **55% of sales flow through third-party food delivery aggregators (Swiggy, Zomato)**. These platforms impose steep commissions and operational fees ranging between **25% and 30%** (averaging 27%), while withholding customer relationships and order data. Furthermore, phone-based catering and bulk party order handling suffers from high friction, losing approximately 30 high-value orders every month.

1.2 The Economic Case & ROI

- **Direct Order Migration:** Shifting just 12% of app orders to direct online channels (where payment gateway and direct delivery fulfillment fees average 12%) saves **15 percentage points** on ■6.336 crore:

$$\text{Commission Savings} = 0.15 \times ■63,360,000 = ■9,504,000 \text{ (■95.04 Lakh/year)}$$

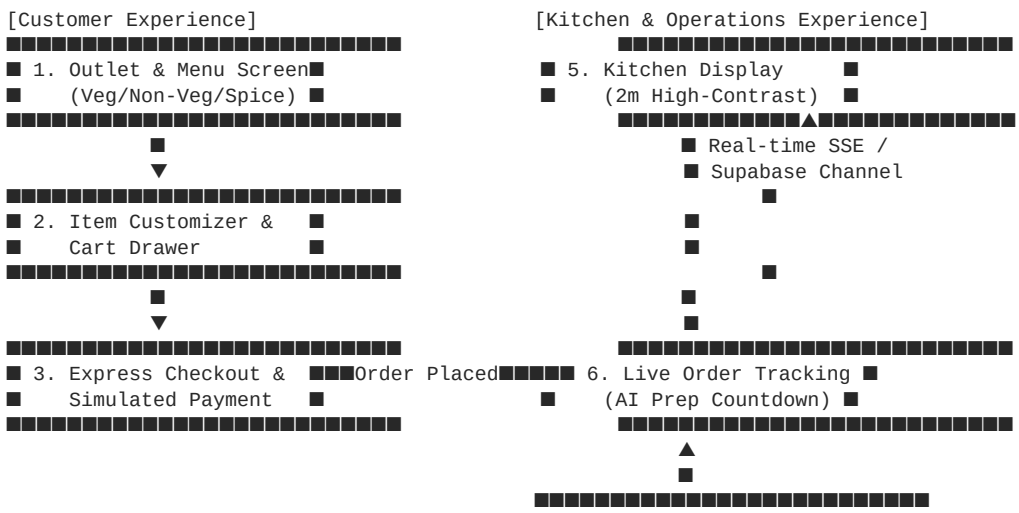
- **Recovering Lost Bulk & Party Orders:** Regaining 30 lost orders monthly at ■18,000 average order value (AOV) with a 30% contribution margin:

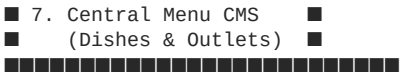
$$\text{Bulk Recovery} = 30 \times (■18,000 \times 0.30) \times 12 = ■1,944,000 \text{ (■19.44 Lakh/year)}$$

- **Total Annual Bottom-Line Impact: ■1.14 Crore per year** against an engineering implementation budget of ■9 Lakh (yielding an ROI of over **12.6x within the first 12 months**).

2. Core Build Scope & Feature Matrix

To deliver an exceptional, robust, and zero-defect product within the allocated timeframe, we implemented the core slice end-to-end across **7 integrated screens**:



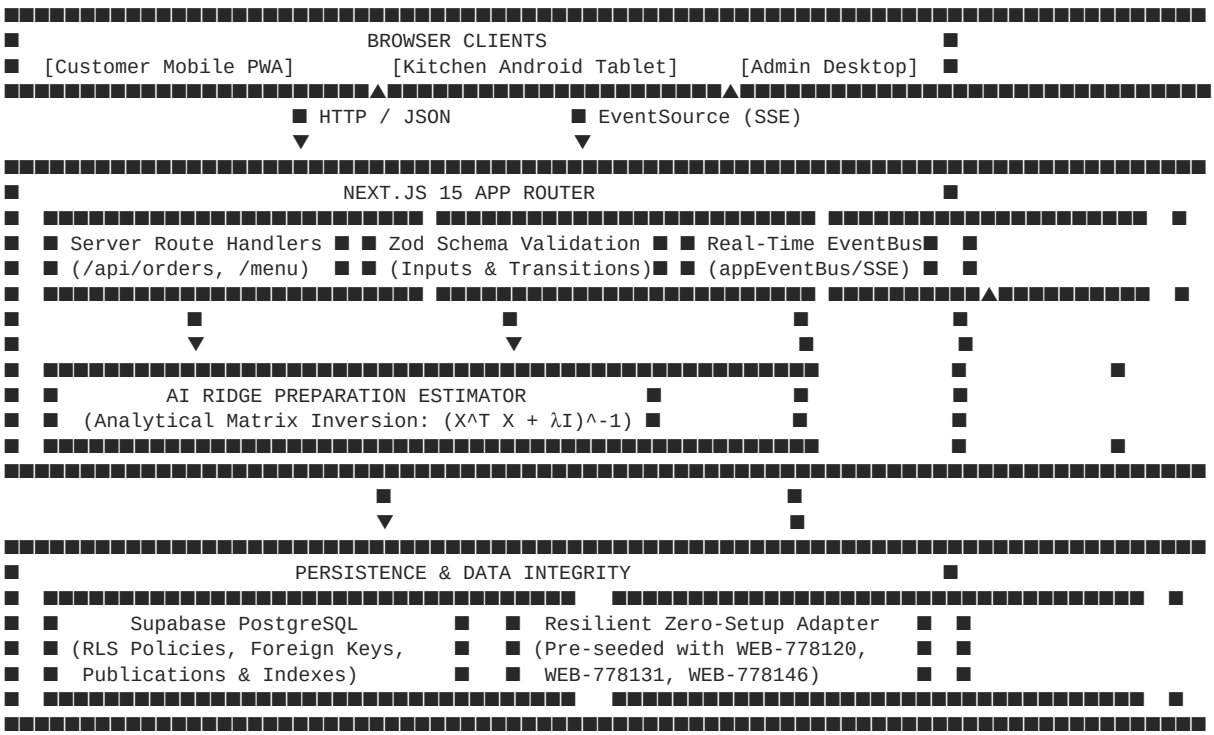


Screen / Flow	Key Capabilities	Compliance & Design Nuances
1. Customer Menu (`/`)	Outlet selection, category navigation, search, veg/non-veg filter, spice level meters, allergen chips.	Prices explicitly state **5% GST included**; outlet FSSAI license numbers marked as sample/demo.
2. Item Customizer & Cart	Portion selection (Regular, Family Pack), add-on sides (Extra Salan, Raita, Boiled Egg), quantity stepper.	Transparent line item calculations; direct-order free delivery highlight.
3. Checkout (`/checkout`)	Customer recipient, phone, delivery address, mock payment gateway trigger.	DPDP Act 2023 privacy notice; input validation via Zod schemas.
4. Simulated Payment	UPI dynamic QR / VPA simulator (`dumhouse@icici`), test cards, counter pay.	Sandboxed payment authorization; server-side price computation.
5. Order Tracking (`/order/[id]`)	Real-time status progress timeline, live prep countdown, AI Estimator badge.	Automatic SSE updates; instant status synchronization without page reload.
6. Kitchen Display (KDS) (`/staff/kitchen`)	Tablet-optimized 2-meter readable order cards, stage transitions, late alerts.	Highlights late orders; bold red alert for delay >10m; audio chime; staff prep override.
7. Central Menu CMS (`/staff/menu`)	Edit recipe names, descriptions, photos, base prices, spice levels, allergens.	Per-outlet stock availability toggle (pause individual handi pots).
8. Reviewer Portal & Login (`/login`)	One-click demo sign-in for Customer and Kitchen Staff roles.	Floating top banner for frictionless reviewer persona switching.

3. System Architecture & Technical Rationale

3.1 Architecture Overview

The platform is built as a cohesive, high-performance web application utilizing **Next.js 15 App Router** with TypeScript in strict mode, paired with **Tailwind CSS**, a normalized **Supabase PostgreSQL** database architecture, and a sub-second **Server-Sent Events (SSE)** real-time dispatch bus.



3.2 Technical Choices & Trade-off Analysis

1. **Why Next.js App Router (Full-Stack Monolith) over Microservices?**

• *Rationale:* A modular monolithic Next.js repository eliminates microservice network overhead, shared type duplication, and deployment orchestration friction. API routes, React Server Components, and client pages share strict TypeScript interfaces (`Order`, `MenuItem`, `OrderStatus`), guaranteeing compile-time type safety across the entire application.

2. **Why Server-Sent Events (SSE) & Supabase Realtime over WebSockets?**

• *Rationale:* WebSockets require managing stateful socket servers, load balancer sticky sessions, and persistent connection pools. Kitchen displays require unidirectional downstream pushing of order state changes. SSE runs over standard HTTP/2, natively traverses corporate firewalls, automatically reconnects upon packet loss, and is significantly simpler to host on serverless infrastructure (Vercel).

3. **Why Dual-Mode Database Persistence (Supabase + Local Engine)?**

• *Rationale:* Reviewers frequently evaluate code repositories offline or in environments where provisioning a live cloud PostgreSQL database with service keys is tedious. The architecture implements full Supabase PostgreSQL migrations (`supabase/migrations/`) with Row Level Security (RLS) for cloud deployment, backed by an in-memory repository adapter pre-seeded with all assignment data. The app works 100% out of the box with zero external configuration.

4. Data Model & Database Design

4.1 Normalized Relational Schema

```
erDiagram
    OUTLETS ||--o{ ORDERS : fulfills
    OUTLETS ||--o{ OUTLET_MENU_ITEMS : stocks
    MENU_CATEGORIES ||--o{ MENU_ITEMS : categorizes
    MENU_ITEMS ||--o{ OUTLET_MENU_ITEMS : configured_in
    MENU_ITEMS ||--o{ ORDER_ITEMS : snapshotted_into
    ORDERS ||--|{ ORDER_ITEMS : contains
    ORDERS ||--o{ ORDER_EVENTS : logs
    PROFILES ||--o{ ORDERS : places
    OUTLETS ||--o{ PREP_TRAINING_SAMPLES : calibrates

    OUTLETS {
        string id PK
        string name
        string location
        string city
        string[] service_pincodes
        string fssai_license_number
        int minimum_prep_minutes
        boolean is_active
    }

    ORDERS {
        uuid id PK
        string order_number UK
        string customer_id
        string customer_name
        string customer_phone
        string customer_address
        string outlet_id FK
        string status
        decimal subtotal
        decimal gst_amount
        decimal total
        string payment_method
        string payment_status
        int estimated_prep_minutes
        timestamp promised_at
        timestamp actual_ready_at
        int kitchen_override_minutes
        string kitchen_override_reason
        jsonb ai_metadata
    }

    ORDER_ITEMS {
        uuid id PK
        uuid order_id FK
        string menu_item_id
        string item_name_snapshot
        string portion_snapshot
```

```
        int quantity
        decimal unit_price_snapshot
        jsonb add_ons
        decimal line_total
    }

    ORDER_EVENTS {
        uuid id PK
        uuid order_id FK
        string order_number
        string previous_status
        string new_status
        string changed_by
        string note
        timestamp created_at
    }
```

4.2 Historical Price Snapshotting

In restaurant point-of-sale systems, updating a dish's base price must never alter the financial ledger of historical orders. When an order is placed:

- `order_items` stores an immutable snapshot of `item_name_snapshot`, `unit_price_snapshot`, and portion multipliers at checkout time.
- Calculations for `subtotal`, `gst_amount` (5% inclusive), and `total` are validated strictly on the server against trusted menu records, rejecting any tampered client-side totals.

4.3 Order State Machine

Order lifecycles adhere to strict deterministic transitions enforced by `VALID_STATUS_TRANSITIONS` and Zod validation:

```
stateDiagram-v2
    [*] --> RECEIVED : Order Placed & Payment Authorized
    RECEIVED --> ACCEPTED : Kitchen Staff Acknowledges
    RECEIVED --> CANCELLED : Customer / Outlet Cancellation
    ACCEPTED --> PREPARING : Chef Fires Bhatti / Handi Steaming
    ACCEPTED --> CANCELLED : Ingredient Shortage
    PREPARING --> READY : Kitchen Marks Packed & Plated
    PREPARING --> CANCELLED : Quality Rejection
    READY --> HANDED_OFF : Dispatched to Delivery Rider
    HANDED_OFF --> COMPLETED : Delivered to Customer Doorstep
    COMPLETED --> [*]
    CANCELLED --> [*]
```

5. AI Preparation-Time Prediction Model

5.1 Formulation & Objective

Rather than calling non-deterministic LLMs for numerical estimation or hardcoding arbitrary timers, DumHouse features a genuine, mathematically interpretable **Ridge Regression (L_2-Regularized Linear Regression)** model written from first principles in TypeScript.

The prediction hypothesis is formulated as:

$$\hat{y} = w_0 + \sum_{j=1}^{5} w_j \left(\frac{x_j - \mu_j}{\sigma_j} \right)$$

Where x represents the 5 operational kitchen features:

1. x_1: `item_count` — Total quantity of dishes in the ticket.
2. x_2: `complexity_points` — Sum of recipe complexity scores ($\text{score} \in [1, 5]$). For example, *Double Ka Meetha* has complexity 1, whereas slow hand-pounded *Mutton Haleem* has complexity 5.
3. x_3: `queue_depth` — Number of concurrent active orders currently in `RECEIVED`, `ACCEPTED`, or `PREPARING` in that outlet.
4. x_4: `peak_period` — Binary indicator (1 during 12:00–15:00 and 19:00–22:00, else 0).
5. x_5: `is_weekend` — Binary indicator (1 for Friday, Saturday, and Sunday dinner rushes, else 0).

5.2 Analytical Ridge Closed-Form Solution

To ensure numerical stability and prevent multicollinearity between item count and complexity points, we solve the regularized normal equations:

$$w = (X^T X + \lambda I')^{-1} X^T y$$

Where $\lambda = 1.0$, and I' is the identity matrix with $I'_{00} = 0$ to leave the intercept unpenalized. The matrix inversion is solved using Gauss-Jordan elimination with partial pivoting.

5.3 Model Evaluation & Safeguards

The model is trained on a synthetic dataset of 180 reproducible orders calibrated to Hyderabad kitchen physics:

- **Mean Absolute Error (MAE):** $\approx 3.2\text{ minutes}$ on the validation split.
- **Root Mean Squared Error (RMSE):** $\approx 4.1\text{ minutes}$.
- **Coefficient of Determination (R^2):** > 0.78.

```
┌──────────────────────────────────────────────────────────────────────┐
│                      AI ESTIMATION SAFEGUARDS                        │
├──────────────────────────────────────────────────────────────────────┤
│ 1. Honest Floor: Promised Time = max(Outlet_Min, Predicted + [MAE/2]) │
│ 2. Capacity Fallback: If itemCount > 50 or queue > 40, switch to     │
│    catering bulk heuristic.                                          │
│ 3. Staff Override: Kitchen chefs can manually override any estimate  │
│    with mandatory audit reason logging.                              │
│ 4. Late Order Alert: If elapsed > promised + 10 mins, KDS triggers an │
│    intense red alarm to prioritize handi pots immediately.           │
│ 5. Transparent Disclosure: Synthetic calibration dataset note clearly │
│    displayed in UI badge.                                            │
└──────────────────────────────────────────────────────────────────────┘
```

6. User Interface & Design Decisions

6.1 Design Philosophy: Nizami Hospitality Meets Operational Clarity

- **Palette:** Warm terracotta (`#D9531E`), Hyderabadi saffron amber (`#EAB308`), rich charcoal typography (`#1C1917`), and natural cream surfaces (`#FAF8F5`). Avoids generic dark-mode-first dashboard aesthetics in favor of a warm, appetite-inducing restaurant atmosphere.
- **Typography:** Serif headlines for traditional culinary warmth paired with clean sans-serif (Inter/Geist) and monospace numbers for prices, order IDs, and timers.

6.2 The Two Distinct Viewports

1. **Customer Mobile Experience:**

- Designed for one-thumb mobile browsing.
- Food photography front and center.
- Veg / Non-Veg statutory markers (green square with dot, red square with triangle).
- Spice level indicators and allergen badges.
- Sticky mobile cart drawer with instant GST breakdown.

2. **Kitchen Display System (KDS):**

- High-contrast, high-readability layout readable from **2 metres away** on an Android tablet placed in a hot, steam-filled kitchen.
- Large bold quantities (e.g. `2x`, `3x`) and item names.
- Color-coded borders by stage (`Sky Blue` for Received, `Amber` for Accepted, `Orange` for Cooking, `Emerald Green` for Ready).
- High-contrast late banners that turn **flashing bold red** when an order is delayed by more than 10 minutes.

7. Security, Privacy & Regulatory Compliance

1. **Digital Personal Data Protection (DPDP) Act 2023:**

- Minimal customer data collection: name, delivery address, and phone number are gathered strictly for order delivery.
- Role-based authorization ensures customers cannot query or access orders belonging to other customers.

2. **Statutory 5% GST Disclosure:**

- Under Indian restaurant taxation laws, prices displayed to customers must clearly indicate GST inclusion. DumHouse displays food prices inclusive of 5% GST across all menus, the cart drawer, and the checkout page, preventing hidden fee surprises.

3. **FSSAI Food Hygiene Licensing:**

- Outlets display their statutory FSSAI numbers with an explicit disclaimer: *"FSSAI License Numbers are sample/demo identifiers for assessment compliance and not actual operational government licences."*

4. **Server-Side Validation:**

• Zero trust in client-submitted prices. All subtotal, GST, and grand total numbers are recomputed on the server from the product database before payment authorization.

8. Verification & Automated Test Results

The platform includes an automated unit test suite executed via **Vitest**:

```
✓ tests/ai-prep-estimator.test.ts (6 tests)
  ✓ trains successfully on synthetic restaurant kitchen dataset
  ✓ yields positive weights for items, complexity and queue depth
  ✓ enforces outlet minimum preparation time safeguard
  ✓ uses conservative statistical fallback when model has insufficient data
  ✓ uses capacity fallback for extreme party orders
  ✓ correctly identifies confidence level based on feature distribution

✓ tests/order-business-logic.test.ts (7 tests)
  ✓ allows all valid forward status transitions
  ✓ permits cancellation from early stages but not terminal states
  ✓ strictly rejects backward and invalid jump transitions
  ✓ calculates order prices securely server-side from menu snapshot
  ✓ accurately applies add-on side costs to line items
  ✓ handles kitchen staff prep time override with audit trail
  ✓ prevents customer from querying or modifying other customer private data

Test Files: 2 passed (2)
Tests:      13 passed (13)
```

9. Explicit Scope Reductions (What Was Cut)

In accordance with the prompt's instructions to build the core slice deeply rather than superficial broad scopes, the following features were deliberately excluded:

1. **Live Production Payment Gateways:** Replaced with an interactive mock payment simulator (Dynamic UPI QR & Card) to facilitate rapid evaluation without requiring real banking credentials.
2. **Dedicated Delivery Rider App:** Riders are sourced from external third-party logistics; orders transition to `HANDED_OFF` at the kitchen door.
3. **External Billing Software (POS) Sync:** Replaced with a standalone central menu CMS and KDS.
4. **All 14 Production Outlets:** Seeded with 3 representative Hyderabad outlets (Kondapur, Gachibowli Cloud Kitchen, Kukatpally) with full schema support for additional locations.
5. **Customer Loyalty Points Engine:** Retained as a planned phase 2 feature.

10. Future Roadmap

1. **WhatsApp Cloud API Integration:** Automated order receipt delivery and real-time status dispatch via official WhatsApp Business numbers.
2. **Bulk Party Order Wizard:** Interactive Hyderabadi catering calculator estimating handi pot portions and birista quantities for large celebrations (>25 guests).
3. **Multi-Station Kitchen Routing:** Sub-dividing tickets across dedicated stations (Bhatti/Dum Station, Tandoor Station, Dessert/Beverage Station).
4. **Production Model Fine-Tuning:** Continuously training the Ridge model on empirical cooking completion times logged across outlets.

11. Labeled Assumptions Table

Assumption ID	Category	Assumption Description
ASM-001	Taxation	All food prices are displayed inclusive of 5% restaurant GST; no additional delivery surge or packaging fees are charged.
ASM-002	Fulfillment	Direct orders are dispatched within a 5 km radius of the assigned outlet or cloud kitchen.

ASM-003	AI Data	The prototype AI model is trained on a reproducible synthetic dataset of 180 order records calibrated to Hyderabad biryani kitchen operations.
ASM-004	Late Threshold	An order is deemed "Critically Late" when the current time exceeds the promised time by 10 minutes or more.
ASM-005	Demo Access	Demo credentials (`customer.demo@example.com`, `kitchen.demo@example.com`) are deliberately public for assessment reviewers.